Opportunity Expiry Predictor

Full Technology Stack & Implementation Specification Report

Project: Opportunity Expiry Predictor	**Runtime Target:** IBM Z / LinuxONE Enterprise Linux
Architecture: Full-Stack Decoupled (FastAPI + React 19)	**Risk Algorithm:** Explainable AI 40-30-20-10 Heuristic
Authentication: JWT + PBKDF2-HMAC-SHA256 (Salted)	**Database:** SQLAlchemy ORM (PostgreSQL & SQLite)

1. Executive Summary & Vision

The **Opportunity Expiry Predictor** is an AI-powered SaaS web application designed for students and job seekers to discover, track, evaluate, and prioritize valuable academic and career milestones (scholarships, internships, competitive examinations, hackathons, fellowships, and job openings). Engineered for **IBM Z and LinuxONE**, the platform provides automated deadline urgency detection, natural language document extraction, and transparent explainable recommendations.

The system replaces fragmented tracking spreadsheets with an intelligent priority heuristic (0 to 100) that alerts candidates to imminent deadlines, excessive documentation requirements, or eligibility gaps, offering prescriptive guidance to submit competitive applications on time.

2. Complete Technology Stack Matrix

Layer	Component	Version / Library	Architectural Purpose & Capability
Frontend Core	React.js + Vite	React 19, Vite 8	High performance SPA with instant HMR and reactive modular state management.
Routing	React Router DOM	v7.x	Client-side routing: /dashboard, /profile, /opportunities, /recommendations, /alerts, /settings, /login, /register.
Iconography	Font Awesome React	@fortawesome/free-solid	Strictly uses FontAwesomeIcon components across navigation, categories, and status indicators.
Styling System	Vanilla CSS3 Tokens	CSS Grid & Flexbox	Clean custom design system: dark navy sidebar (#172D50), primary blue (#2563EB), pastel stat cards, and responsive media queries.
Visualization	SVG Progress & Recharts	SVG / Recharts	Radial progress gauge for Profile Match and multi-dimensional profile alignment breakdowns.
Backend API	Python FastAPI	FastAPI 0.141+	Asynchronous ASGI REST framework with automated OpenAPI / Swagger interactive documentation.
ASGI Server	Uvicorn	Uvicorn 0.52+	High throughput asynchronous web server supporting hot-reloading and enterprise hosting.
Data Validation	Pydantic	Pydantic v2.13+	Type-safe request/response serialization, schema validation, and error reporting.
ORM & Database	SQLAlchemy ORM	SQLAlchemy 2.0+	Normalized models supporting PostgreSQL for production and zero-config SQLite for local development.
Security & Auth	JWT + PBKDF2 HMAC	python-jose, hashlib	Stateless JSON Web Tokens with cryptographically salted PBKDF2-HMAC-SHA256 password security.
Document AI	PyMuPDF	PyMuPDF 1.28.2 (fitz)	Extracts text, deadlines, eligibility criteria, and required documents from uploaded PDF circulars.
Web Scraping / NLP	BeautifulSoup4 + Regex	BS4 4.15+, Requests	Parses official announcements, extracts date patterns, and normalizes unstructured circular notices.
Target Runtime	IBM Z / LinuxONE	s390x Enterprise Linux	High security, continuous availability, and enterprise transaction resilience.

3. Detailed Implementation Breakdown

The following modules represent the complete implementation delivered across the frontend and backend:

A. Visual Design System & Reference Dashboard Layout

- **Precision Palette:** Dark navy sidebar (#172D50), primary blue (#2563EB), light canvas (#F4F7FC), white card surfaces (#FFFFFF), borders (#E2E8F0).
- **Typography:** Inter sans-serif typeface configured with exact hierarchical scale (Headings: 22px, Section titles: 16px, Table/Meta: 11-13px).
- **Fixed Dark Navy Sidebar:** Fixed 190px sidebar with brand mark, active nav pill in bright blue, Font Awesome icons, and IBM Z footer panel: *'Deployed on IBM Z for a smarter tomorrow'*.
- **Navigation Items:** Dashboard (faHouse), My Profile (faUser), Opportunities (faBriefcase), Recommendations (faStar), Alerts (faBell), Settings (faGear), and Log Out (faRightFromBracket).

B. Header & Live Database-Backed Stat Cards

- **Dashboard Header:** Dynamic greeting (*'Hello, [User Name]!'*), search input with faMagnifyingGlass, user avatar, and user menu dropdown.
- **4 Pastel Stat Cards:** Calculated live from backend database records (Total Opportunities: soft blue #EAF3FF; Expiring Soon within 7d: soft yellow #FFF7E6; High Risk: soft red #FFF0F0; Recommended for You: soft green #E9FAF1).
- **Top Urgent Opportunities:** Displays top time-sensitive rows with category icon, deadline date, days remaining badge, risk level badge, match %, and 'View' modal button.
- **All Opportunities Data Table:** Live multi-criteria filtering (search title/org, category dropdown, risk level dropdown), sortable columns (deadline, risk score, match %), pagination, and empty/loading states.

C. Right-Side Intelligence Panel

- **Your Profile Match:** Radial circular progress chart showing overall profile match % with 4 dimensional indicators (Education 94%, Skills 70%, Interests 90%, Career Goals 85%).
- **Upcoming Deadlines:** Next 4 impending opportunities with urgency badges (Red for critical <=2d, Amber for approaching <=7d, Green for normal).
- **Quick Insights:** Dynamic AI takeaways generated from database state with direct 'View Recommendations' button.
- **Smart Alerts:** Soft-blue card summarizing recent notifications with unread badges linking to the Alert Center.

D. Explainable AI Risk Scoring Engine (40-30-20-10 Heuristic)

Implemented in backend/risk_engine.py, this engine computes an explainable risk score (0 to 100) identifying opportunities at high risk of being missed:

Weight	Evaluation Factor	Algorithmic Rules & Contributing Criteria
40%	**Deadline Urgency**	Compares remaining days against preparation days. Days <= 1 yields 100/100; <= 2 yields 95/100 (Critical); <= 7 yields 65/100; prep time exceeding remaining time adds +20 penalty.
30%	**Eligibility Mismatch**	Compares user's CGPA, degree branch, and technical skill tags against opportunity minimum criteria.
20%	**Preparation Burden**	Quantifies documentation difficulty based on required documents (transcripts, SOP essays, recommendation letters, pitch decks).
10%	**Missing Info / Verification**	Penalizes unverified deadlines (+75) or missing application URLs (+20) requiring manual confirmation.

- **Opportunity Detail Modal:** Clicking 'View' displays transparent factor breakdowns (e.g. *'Critical urgency: Application deadline is in 2 days'*), suggested actions, required documents, and official portal link.

E. Automatic NLP Document & Notice Extraction

- **PDF Notice Extraction:** Uses PyMuPDF to extract text from official uploaded PDF circulars.

- **Date & Deadline NLP:** Intelligently detects application cutoff dates across unstructured phrasing (e.g. *'Last Date of Application: 30 November 2026'*).
- **Webpage Parsing:** Employs BeautifulSoup4 to extract clean text from authorized university announcement pages.
- **Interactive Extractor Modal:** Located on the Opportunities page; allows pasting text or uploading PDFs to preview extracted fields, confidence scores, and save directly to the database.

F. Profile Management & Dynamic Recalculation

- **Profile Management (/profile):** Candidates manage degree, university, CGPA, technical skills, areas of interest, preferred categories, and career goals.
- **Live Profile Completion Bar:** Tracks completeness percentage in real-time.
- **Automatic Reactive Recalculation:** Updating and saving profile immediately triggers background recalculation of match percentages and risk assessments across all opportunities.

G. Authentication, Sign-in, Registration & Logout

- **Registration Page (/register):** Account creation with degree/branch selection, password validation, and automatic onboarding.
- **Login Page (/login, /signin):** Clean auth view with password show/hide toggle and 'Remember me' support.
- **1-Click Demo Student Access:** Instant login as Alex Chen (student@university.edu) pre-seeded with active opportunities and notifications.
- **Logout Option:** Seamlessly accessible from both the Header user dropdown and Left Sidebar, clearing tokens and returning to sign-in.

H. Recommendations, Alerts & Settings Pages

- **Recommendations (/recommendations):** Ranked opportunities with explainable reasoning (*'Why this is recommended for you'*) and dismiss actions.
- **Alert Center (/alerts):** In-app notification center tracking approaching deadlines with bulk 'Mark all as read'.
- **Settings (/settings):** Configurable reminder schedule (7d, 3d, 1d, 0d) and IBM Z / LinuxONE telemetry status.

4. Relational Database Schema & REST API Endpoints

The backend database contains 11 normalized SQLAlchemy models with complete foreign keys and cascade rules:

Entity Model	Key Fields	Description & Relationships
User	id, email, hashed_password, full_name, is_active	Core authentication model linked to Profile, Bookmarks, and Notifications.
UserProfile	id, user_id, educational_qualification, degree_branch, cgpa, skills	Academic standing and technical profile used for dynamic matching.
Opportunity	id, title, organization, category, deadline, min_cgpa, is_verified, status	Master opportunity catalog with verification status and prep requirements.
EligibilityCriteria	id, opportunity_id, criteria_type, description, required_value	Fine-grained eligibility rules (degree, CGPA, year, skills).
RequiredDocument	id, opportunity_id, document_name, is_mandatory	Mandatory and optional documents evaluated for preparation burden.
RiskAssessment	id, user_id, opportunity_id, risk_score, risk_level, reasons_json	Stored explainable 4-factor risk assessment scores.
OpportunityRecommendation	id, user_id, opportunity_id, match_percentage, reason	Ranked recommendations with plain-English reasoning.
Notification	id, user_id, opportunity_id, title, message, type, is_read	Deadline warnings and recommendation alerts.
UserBookmark	id, user_id, opportunity_id, created_at	Saved opportunities tracked by individual candidates.
OpportunitySource	id, opportunity_id, source_type, source_url, raw_content, confidence	Audit log of extracted PDF notices and web circulars.

FastAPI Endpoints Summary

- POST /auth/register & POST /auth/login — User registration and JWT authentication
- GET /users/me & PUT /users/me — Fetch/update profile with dynamic score recalculation
- GET /opportunities — Filtered opportunity catalog (search, category, risk level, pagination)
- GET /opportunities/urgent — Top urgent opportunities prioritized by days left and risk score
- GET /opportunities/recommended — Personalized opportunity recommendations with transparent reasoning
- GET /opportunities/{id} — Detailed opportunity breakdown with documents and criteria
- POST /opportunities/upload — Document & text extraction via PyMuPDF / BeautifulSoup4
- GET /analytics/dashboard — Live database dashboard statistics (Total, Expiring, High Risk, Recommended)
- GET /notifications & PATCH /notifications/{id}/read — Notification management
- POST /bookmarks/{id} & DELETE /bookmarks/{id} — Opportunity bookmarking

5. Local Execution Instructions

Start Backend API: cd backend && python -m uvicorn main:app --host 127.0.0.1 --port 8000 --reload

Start Frontend App: cd frontend && npm run dev -- --host 127.0.0.1 --port 5173

Interactive Swagger Docs: http://127.0.0.1:8000/docs | **Dashboard:** http://127.0.0.1:5173/

Summary: *The Opportunity Expiry Predictor is a production-quality, data-backed full-stack web application faithfully matching the reference design and deployed ready for IBM Z / LinuxONE enterprise infrastructure.*